US 20240394543A1

(19) United States

(12) Patent Application Publication (10) Pub. No.: US 2024/0394543 A1

Mathew et al. (43) Pub. Date: Nov. 28, 2024

(54) **PARAMETRIC POWER-OF-2 CLIPPING ACTIVATIONS FOR QUANTIZATION FOR CONVOLUTIONAL NEURAL NETWORKS**

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(21) Appl. No.: **18/795,565**

(22) Filed: **Aug. 6, 2024**

Related U.S. Application Data

(63) Continuation of application No. 17/117,271, filed on Dec. 10, 2020, now Pat. No. 12,099,930.

(30) **Foreign Application Priority Data**

Dec. 12, 2019 (IN) 201941051538

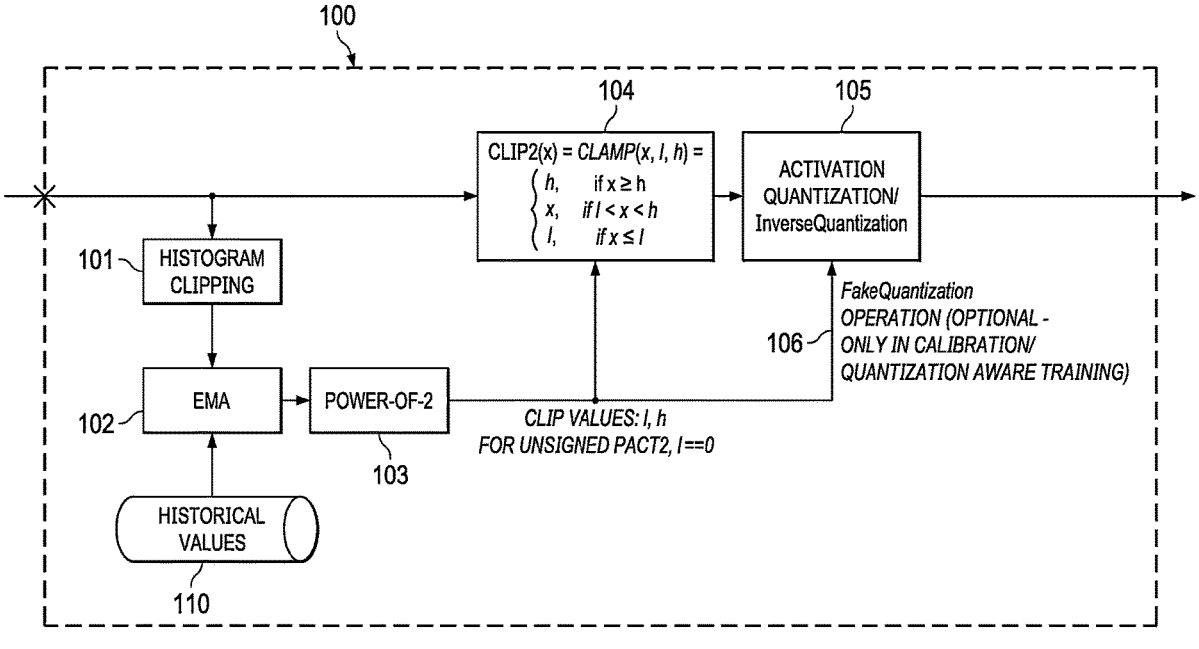
Publication Classification

(51) **Int. Cl.**

G06N 3/084 (2006.01)
G06N 3/04 (2006.01)
G06N 3/048 (2006.01)

(52) **U.S. Cl.**

CPC ***G06N 3/084*** (2013.01); ***G06N 3/04*** (2013.01); ***G06N 3/048*** (2023.01)

(57) **ABSTRACT**

In an example, a method includes executing, using one or more processors, a power-of-2 parametric activation (PACT2) function to quantize a set of data. The executing of the PACT2 function includes determining a distribution for the set of data; discarding a portion of the data corresponding to a tail of the distribution to form a remaining set of data; estimating a maximum value of the remaining set of data; determining a new maximum value of the remaining set of data using a moving average and at least one historical value of at least one prior remaining set of data; determining a clipping value by expanding the new maximum value to a nearest power of two value; and quantizing the set of data using the clipping value to form a quantized set of data.